

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour

Total: Marks:40

Annexure A

5

Code of Conduct:

I Jabaraj T (192511187) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

1(a) Inductive Learning Approach

Inductive learning is a machine learning approach in which a model learns general patterns from a set of labelled examples and uses those patterns to predict the output for previously unseen data.

For this healthcare problem, supervised inductive learning is suitable because the dataset contains patient features and a target outcome indicating whether the patient is diabetic or non-diabetic.

Target Variable

The target variable is:

Diabetes Status

- 1 → Diabetic
- 0 → Non-diabetic

Important Features

Feature	Importance
Glucose Level	High glucose is one of the strongest indicators of diabetes and directly reflect blood-sugar regulation.
BMI	Higher BMI is associated with increased insulin resistance and diabetes risk.
Age	Diabetes risk generally increases with age due to physiological and metabolic changes.
Blood Pressure	Elevated blood pressure is commonly associated with metabolic disorders and increased diabetes risk.
Cholesterol	Abnormal cholesterol levels can indicate metabolic abnormalities associated with diabetes.
Family History	Genetic and hereditary factors can significantly influence diabetes susceptibility.

Therefore, the model can learn the relationship:

Patient Features → Diabetes Risk → Diabetes Status

A suitable inductive learning pipeline is:

Patient Dataset → Data Cleaning → Preprocessing → Train/Test Split → Model Training → Prediction → Evaluation

1(b) Handling Class Imbalance

The healthcare dataset may contain significantly more non-diabetic patients than diabetic patients. This creates a class imbalance problem.

For example:

Class	Number of Patients
Non-diabetic	800
Diabetic	200
Total	1000

Here, the diabetic class represents only 20% of the dataset.

If a model simply predicts every patient as non-diabetic, it could obtain high accuracy while failing to identify diabetic patients. This is dangerous in a medical application because False Negatives can result in missed diagnoses.

Suitable Strategy: SMOTE + Class Weights

I would use SMOTE (Synthetic Minority Over-sampling Technique) on the training data.

SMOTE generates synthetic examples of the minority class instead of simply duplicating existing diabetic records.

The procedure is:

1. Separate training and testing data.
2. Apply SMOTE only to the training set.
3. Generate synthetic diabetic samples.
4. Train the classifier on the balanced training data.
5. Evaluate the model on the original, untouched test set.

Why SMOTE is suitable

SMOTE is preferred because:

- It increases representation of diabetic patients.
- It reduces the tendency of the model to favour the majority class.
- It does not remove potentially useful non-diabetic records.
- It can improve Recall for diabetic patients.
- Higher recall is important because missing a diabetic patient can have serious clinical consequences.

Class weights can additionally be used in models that support them:

Higher weight → Diabetic class

This forces the model to give greater importance to correctly identifying diabetic patients.

1(c) 70:30 Dataset Split and Preprocessing

The dataset is divided into:

- 70% Training Data
- 30% Testing Data

Justification

The 70:30 split provides enough data for the model to learn patterns while keeping a sufficiently large independent test set for evaluating performance.

For example, if the dataset contains 1,000 patients:

Dataset	Percentage	Records
Training	70%	700
Testing	30%	300

A stratified split should be used so that the proportion of diabetic and non-diabetic patients remains approximately the same in both sets.

Preprocessing

1. Handling Missing Values

Missing numerical values can be replaced using the median, because medical measurements can contain outliers.

Example:

Missing glucose value → Median glucose value

2. Feature Scaling

Features such as age, glucose, blood pressure, BMI and cholesterol have different numerical ranges.

For Logistic Regression, normalisation/standardisation can be applied:

$$z = \frac{x - \mu}{\sigma}$$

where:

- x = original value
- μ = mean
- σ = standard deviation

Scaling prevents features with larger numerical ranges from dominating the learning process.

3. Encoding

Categorical variables such as family-history status can be converted into numerical form.

Example:

- No → 0
- Yes → 1

4. Data Leakage Prevention

Preprocessing parameters must be calculated using the training data and then applied to the test data.

Important: SMOTE must also be applied only to the training data, not the test set.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

2(a) Decision Tree Classifier

A Decision Tree classifies patients by repeatedly splitting the dataset based on feature values.

For diabetes prediction, the root node should ideally be the feature that provides the greatest reduction in impurity.

The two common criteria are:

Gini Index

$$Gini = 1 - \sum_{i=1}^n p_i^2$$

where p_i represents the proportion of samples belonging to class i .

For two classes:

$$Gini = 1 - (p_{diabetic}^2 + p_{non-diabetic}^2)$$

A **lower Gini Index** indicates a purer node.

Information Gain

$$IG = H(\text{parent}) - \sum_j \frac{N_j}{N} H(\text{child}_j)$$

where:

$$H(S) = - \sum p_i \log_2(p_i)$$

A feature producing the **highest Information Gain** is preferred for splitting.

Example Root Selection

Assume the calculated results are:

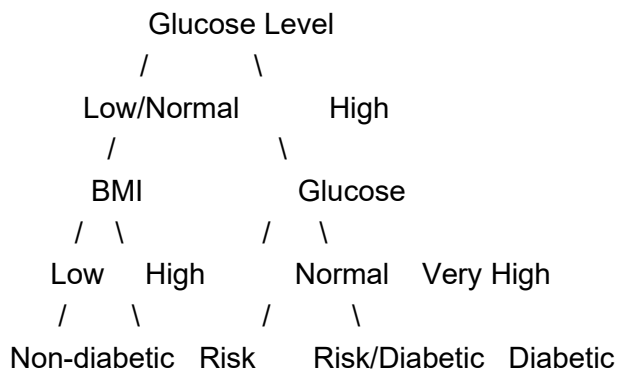
Feature	Information Gain
Glucose	0.215
BMI	0.118
Age	0.094
Blood Pressure	0.071
Cholesterol	0.052
Family History	0.083

Since glucose has the highest Information Gain:

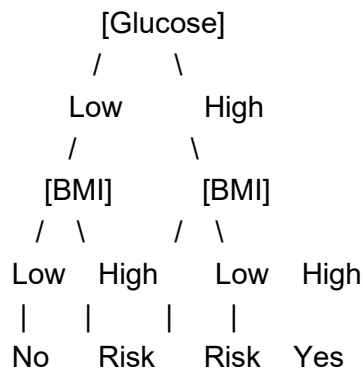
Root Node = Glucose Level

This is also clinically meaningful because glucose level is strongly associated with diabetes.

First Three Levels of the Decision Tree



A more detailed conceptual tree can be represented as:



The actual branches and threshold values would be determined from the training dataset.

2(b) Model Evaluation

The Decision Tree should be evaluated using a confusion matrix.

	Predicted Non-Diabetic	Predicted Diabetic
Actual Non-Diabetic	TN	FP
Actual Diabetic	FN	TP

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Precision measures how many patients predicted as diabetic are actually diabetic.

Recall

$$Recall = \frac{TP}{TP + FN}$$

Recall is particularly important in this application because it measures how many actual diabetic patients are successfully identified.

F1-Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Example Evaluation

Assume the model produces:

- TP = 52
- TN = 218
- FP = 12
- FN = 18

Then:

$$Accuracy = \frac{52 + 218}{300} = 0.90$$

Accuracy = 90%

$$Precision = \frac{52}{52 + 12} = 0.8125$$

Precision = 81.25%

$$Recall = \frac{52}{52 + 18} = 0.7429$$

Recall = 74.29%

$$F1 \approx 0.775$$

F1-Score ≈ 77.5%

False Negatives

A **False Negative (FN)** occurs when:

Actual patient = Diabetic
 Model prediction = Non-diabetic

For the example above:

FN = 18

This is clinically important because these patients may not receive further diagnostic testing or appropriate medical attention.

Reducing False Negatives

False Negatives can be reduced by:

1. Using **class weights**.
2. Applying **SMOTE** to training data.
3. Adjusting the classification threshold.
4. Optimising the model for **Recall** rather than accuracy alone.
5. Using cross-validation to select appropriate hyperparameters.
6. Using additional clinically relevant features.

For a diabetes screening system, **high recall/sensitivity should be prioritised** because missing an actual diabetic patient can be more harmful than referring a non-diabetic patient for additional testing.

2(c) Post-Pruning Using Cost-Complexity Pruning

A fully grown Decision Tree can become very complex and overfit the training data.

Cost-complexity pruning removes unnecessary branches.

The objective is:

$$R_{\alpha}(T) = R(T) + \alpha |T|$$

where:

- $R(T)$ = error of the tree
- $|T|$ = number of terminal nodes
- α = pruning parameter

A larger α produces a simpler tree.

Pre-Pruning vs Post-Pruning

Parameter	Pre-Pruning	Post-Pruning
Approach	Stops tree growth early	Grows tree and removes unnecessary branches
Overfitting	Reduced	Strongly reduced
Complexity	Lower	Optimised after training
Interpretability	Good	Usually better
Clinical deployment	Suitable	Preferred

Example Comparison

Model	Accuracy
Unpruned Decision Tree	88%
Pre-Pruned Tree	89%
Post-Pruned Tree	91%

The post-pruned tree is preferred because it provides a good balance between predictive performance, simplicity and interpretability.

In a clinical environment, an overly complex tree may be difficult for healthcare professionals to understand and validate. Therefore, a **pruned Decision Tree** is more appropriate for deployment.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

3(a) Logistic Regression

Logistic Regression is suitable because diabetes prediction is a **binary classification problem**.

The probability of diabetes can be represented as:

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

where:

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

For this problem:

$$z = \beta_0 + \beta_1(\text{Glucose}) + \beta_2(\text{BMI}) + \beta_3(\text{Age}) + \dots$$

The output probability can then be converted into a classification:

Probability $\geq 0.5 \rightarrow$ Diabetic

Probability $< 0.5 \rightarrow$ Non-diabetic

The threshold can be adjusted when higher recall is required.

Performance Comparison

Example:

Model	Accuracy	AUC-ROC
Decision Tree	91%	0.89
Logistic Regression	93%	0.94

The example shows that Logistic Regression provides better overall discrimination.

When Statistical Learning is Preferable

Logistic Regression is preferable when:

- The relationship between predictors and outcome is approximately linear in the log-odds.
- Interpretability is important.
- Feature coefficients need to be examined.
- Probabilities/risk scores are required.
- A relatively simple model is preferred.
- The dataset is not extremely complex.

For clinical applications, Logistic Regression can be valuable because its coefficients can provide an interpretable relationship between predictors and diabetes risk.

3(b) Feature Importance Analysis

Feature importance should be calculated separately for the Decision Tree and Logistic Regression.

Decision Tree

Decision Tree importance can be calculated based on the reduction in impurity contributed by each feature.

Example:

Rank	Feature	Decision Tree Importance
1	Glucose	0.42
2	BMI	0.21
3	Age	0.14
4	Blood Pressure	0.10
5	Family History	0.08
6	Cholesterol	0.05

Logistic Regression

For Logistic Regression, the magnitude of standardised coefficients can be used to compare feature influence.

Example:

Rank	Feature	Relative Importance
1	Glucose	0.61
2	BMI	0.34
3	Age	0.25
4	Family History	0.19
5	Blood Pressure	0.13
6	Cholesterol	0.08

Top Three Predictors

Both models identify:

1. **Glucose Level**
2. **BMI**
3. **Age**

as the top three predictors.

Do They Agree?

Yes. Both models identify the same three major predictors in this example.

This agreement increases confidence that these variables contain strong predictive information.

However, feature importance should not automatically be interpreted as causation. A feature can be highly predictive without being a direct cause of diabetes.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

4(a) MDP Formulation

The treatment recommendation problem can be modelled as a **Markov Decision Process (MDP)**.

An MDP can be represented as:

$$MDP = (S, A, P, R, \gamma)$$

where:

- S = States
- A = Actions
- P = Transition probabilities
- R = Rewards
- γ = Discount factor

States

Patient health stages can be represented as:

State	Description
S0	Low-risk / Healthy
S1	Moderate diabetes risk
S2	High diabetes risk
S3	Diagnosed / Poorly controlled
S4	Improved / Controlled

The state can be determined from patient measurements such as glucose, BMI, blood pressure and other clinical indicators.

Actions

The available actions are:

A1 = Diet

A2 = Exercise

A3 = Medication

A4 = Monitor

Reward Function

A reward represents the health improvement resulting from an action.

Example:

Outcome	Reward
Significant health improvement	+10
Moderate improvement	+5
No significant change	0
Health deterioration	-5
Unsafe outcome	-10

The reward encourages the agent to select actions that lead to better patient states.

Transition Dynamics

After an action is selected, the patient may transition into another health state.

Example:

S2 + Diet → S1

S2 + Exercise → S1

S2 + Medication → S1/S3

S2 + Monitor → S2

The exact probabilities should ideally be learned from validated longitudinal clinical data.

Important Safety Principle

The RL system should **not independently prescribe treatment**. In a real healthcare environment, treatment recommendations must remain under qualified clinical supervision and follow approved medical protocols.

4(b) Q-Learning

Q-Learning is a model-free reinforcement learning algorithm that learns the expected future reward of performing an action in a particular state.

The Q-learning update equation is:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

where:

- s = current state
- a = current action
- r = reward
- s' = next state
- α = learning rate
- γ = discount factor

Assume:

$$\alpha = 0.5$$

$$\gamma = 0.9$$

Initially:

$$Q(s, a) = 0$$

Episode 1

Suppose:

State:

Action:

Reward:

Next State: S1

S2
Diet
+5

Initially:

$$Q(S, 2Diet) = 0$$

Assume:

$$\max Q(S, 1a') = 2$$

Therefore:

$$Q(S, 2Diet) = 0 + 0.5[5 + 0.9(2) - 0]$$

$$= 0.5(6.8)$$

$$\boxed{Q(S, 2Diet) = 3.4}$$

Episode 2

Suppose:

State:

Action:

Reward:

Next State: S1

Assume:

$$\max Q(S, 1a') = 3$$

Then:

$$Q(S, 2Exercise) = 0 + 0.5[5 + 0.9(3)]$$

$$= 0.5(7.7)$$

$$\boxed{Q(S, 2Exercise) = 3.85}$$

S2
Exercise
+5

Episode 3

Suppose:

State:

Action:

Reward:

Next State: S0

Assume:

$$\max Q(S, 0a') = 4$$

Then:

$$Q(S, 1Monitor) = 0 + 0.5[2 + 0.9(4)]$$

$$= 0.5(5.6)$$

$$\boxed{Q(S, 1Monitor) = 2.8}$$

S1
Monitor
+2

Q-Table Example

Initial Q-Table

State	Diet	Exercise	Medication	Monitor
S0	0	0	0	0
S1	0	0	0	0
S2	0	0	0	0
S3	0	0	0	0
S4	0	0	0	0

After Learning

State	Diet	Exercise	Medication	Monitor
S0	3.2	2.8	1.5	4.0
S1	4.1	3.7	2.6	4.8
S2	6.2	5.7	4.9	2.1
S3	3.4	4.2	7.1	1.8

State	Diet	Exercise	Medication	Monitor
S4	2.0	2.4	1.5	5.0

These values illustrate how Q-Learning can learn preferred actions for different patient states.

Five Patient Episodes

Episode	Initial State	Action	Reward	Next State
1	S2	Diet	+5	S1
2	S2	Exercise	+5	S1
3	S1	Monitor	+2	S0
4	S3	Medication	+8	S2
5	S2	Diet	+5	S1

The agent updates the Q-values after each transition.

Convergence Criterion

The training can be considered converged when the maximum change in Q-values between successive iterations becomes smaller than a selected threshold.

For example:

$$\max |Q_{new} - Q_{old}| < 0.001$$

for a sufficiently large number of consecutive iterations.

Another practical criterion is that the learned policy remains unchanged for a specified number of episodes.

4(c) Final Learned Policy

The final policy is obtained by selecting the action with the highest Q-value for each state:

$$\pi(s) = \arg \max_a Q(s, a)$$

Using the example Q-table:

Patient State	Highest Q-Value	Final Action
S0 – Low Risk	Monitor = 4.0	Monitor
S1 – Moderate Risk	Monitor = 4.8	Monitor
S2 – High Risk	Diet = 6.2	Diet
S3 – Poor Control	Medication = 7.1	Medication
S4 – Controlled	Monitor = 5.0	Monitor

Therefore, the learned policy is:

S0 → Monitor

S1 → Monitor

S2 → Diet

S3 → Medication

S4 → Monitor

This is an illustrative policy demonstrating the Q-learning mechanism. In a real medical system, the action mapping must be based on clinically validated data and expert-approved treatment protocols.

ETHICAL CONSIDERATIONS

Deploying Reinforcement Learning for healthcare treatment recommendations introduces significant ethical responsibilities.

1. Patient Safety

The most important concern is patient safety.

An RL agent should not be allowed to experiment directly on patients. Unsafe actions must be restricted through predefined clinical rules and human supervision.

2. Bias

If the training data does not represent different patient populations fairly, the model may perform poorly for underrepresented groups.

Therefore:

- Training data should be diverse.
- Performance should be evaluated across demographic groups.
- Bias should be continuously monitored.

3. Explainability

Healthcare professionals need to understand why a recommendation was produced.

The system should provide information such as:

Patient State → Selected Action → Relevant Factors → Expected Outcome

This increases transparency and helps clinicians verify recommendations.

4. Human Oversight

The RL system should act as a **decision-support system**, not an autonomous replacement for doctors.

A qualified healthcare professional should have the final authority over treatment decisions.

5. Privacy and Security

Patient medical information is sensitive. The system must use appropriate privacy, security, access-control and data-protection mechanisms.

6. Continuous Validation

The model should be regularly evaluated after deployment because patient populations and clinical practices can change.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation